SYNEQ

INDEPENDENT SYSTEMS STUDIO

CONCEPT / UX / TECHNICAL BLUEPRINT

Useful systems. Real-world momentum.

A practical guide to the SYNEQ studio and its two product concepts:

UniTrade / OrbitX

FROM THE FIRST IDEA TO A RESPONSIBLE, TESTABLE WEB EXPERIENCE

Version 1.0 | October 2026

HTML / CSS / VANILLA JAVASCRIPT

START HERE

At a glance

SYNEQ is presented here as a product and systems studio focused on practical digital experiences. The website introduces two distinct product directions: UniTrade, a campus circular marketplace, and OrbitX, an AI-driven campus fitness and health ecosystem. This guide explains the problem framing, feature concepts, website layout, technical choices, browser-side interactions, important boundaries, and a path from a front-end prototype to a production system.

AREA	WHAT IT COVERS
SYNEQ studio	Shared purpose, principles, contributors, and connected product direction.
UniTrade	Campus-local exchange: buy, sell, rent, borrow, barter, donate, and find items nearby.
OrbitX	Training plans, nutrition awareness, activity, sleep, recovery, and sustainable routines.
Website	Product suite, capability tabs, impact framework, About, team, field notes, search, theme, tour, screenshots, exports, and reports.
Technical model	Three static source files, localStorage preferences, browser-side export helpers, and a companion PDF guide.

DOCUMENT MAP

01. The SYNEQ objective and design principles
02. Product concept: UniTrade
03. Product concept: OrbitX
04. Website information architecture and interaction design
05. Front-end architecture and how to run the files
06. Browser storage, exports, screenshots, and PDF output
07. Accessibility, motion, privacy, and safety boundaries
08. From prototype to production: roadmap and validation

Important scope note: This document describes the intended product concepts and the front-end website package. It does not claim the linked product applications have every feature listed, nor that SYNEQ has a production backend, verified user accounts, verified metrics, or deployed AI services.

01 / THE STUDIO

The SYNEQ objective

Build digital products that turn everyday friction into clearer choices, healthier habits, and shared value.

PRINCIPLE	HOW IT AFFECTS PRODUCT DESIGN
Useful over noisy	Prioritize clear, repeatable actions over features that only look impressive.
Transparent by default	Explain estimates and suggestions; label demo data and describe assumptions.
Built for real constraints	Account for student budgets, schedules, shared spaces, low-friction access, and accessibility.
Better with feedback	Treat the product as an evolving system; measure actual outcomes and refine the experience.

Why two products?

UniTrade and OrbitX address different but related pressures in student life. UniTrade looks at material access and the value of sharing; OrbitX looks at physical wellbeing and the routines that support it. Both begin with practical constraints rather than technology for its own sake.

Brand and positioning

The SYNEQ website uses a dark, high-contrast visual language with an orange-red highlight, cyan and violet accents, animated geometry, compact technical labels, and spacious editorial typography. The goal is an independent systems-studio identity rather than a generic storefront or hackathon landing page.

02 / PRODUCT CONCEPT

UniTrade: Campus Circular Marketplace

Live product website: https://unitradeneo.vercel.app/

The problem

At the start of a semester, students may need books, notes, calculators, drafting boards, lab coats, cycles, mattresses, coolers, furniture, or other hostel essentials. Buying everything new can be expensive when some items are only needed for a short period. At the same time, graduating students may leave usable belongings behind. General-purpose marketplaces can make it harder to find nearby items or arrange a convenient campus handover.

The solution direction

UniTrade is conceived as a college-focused marketplace that supports peer-to-peer exchange inside the campus community. It extends beyond buying and selling by making rental, borrowing, barter, free pickup, donations, and demand requests part of the same discovery experience.

Core capability map

CAPABILITY	PURPOSE	EXAMPLE
Campus-gated access	Keep discovery relevant to a known college or campus community.	Approved institutional email or student-ID verification in a production implementation.
Categorized listings	Make essentials easier to browse and compare.	Books, electronics, dorm essentials, transportation, lab equipment, freebies, and digital resources.
Flexible exchange modes	Match the transaction to how long an item is needed.	Buy, sell, rent by day or week, borrow, barter, donate, or collect for free.
Looking For requests	Let sellers see real demand.	“Looking for an engineering drawing board near Block B before tomorrow.”
Fair-price estimate	Offer a transparent starting range.	Use original price, condition, age, brand, category, and demand signals.
Lost and Found suggestions	Rank possible matches between lost and found descriptions.	Compare item type, colour, brand, location, date, and wording; require human confirmation.

Extensions that create a campus-specific experience

- **Starter Kit bundles:** group textbooks, lab coats, drawing instruments, and other course items into a combined offer with explicit savings.
- **Campus Karma and zero-rupee drops:** recognize useful donations without rewarding spam or artificial activity.
- **Roommate Split-Buy:** show proposed shares, contribution amounts, participant status, and ownership notes. Actual payments must use a compliant provider.
- **Digital Notes Vault:** organize authorized study resources with subject tags, previews, ratings, and clear copyright rules.
- **Vacation Hold:** document storage duration, handover, condition, damage reporting, and cancellation rules.
- **Legacy items and end-of-semester Fire Sale:** make reuse history and time-limited clearance easier to understand.

A sample student journey

A first-year student opens UniTrade before moving into a hostel. They search for a mattress, filter to a nearby block, compare condition and price, and contact the seller without making a phone number public. If no suitable listing exists, they publish a

budget-limited Looking For request. A senior sees the request and offers a used item for pickup. The student checks the condition at a public campus handover point and both parties confirm the exchange.

Suggested MVP sequence

- Start with college selection, demo account, listing creation, search, category filters, and buy/rent/free modes.
- Add Looking For requests and clear seller/buyer contact workflows.
- Introduce donation/Karma and a simple impact dashboard with explicit estimate labels.
- Then add lost-and-found ranking, starter bundles, roommate split-buy simulation, and document sharing with moderation.
- Only after abuse, safety, and data workflows are reliable should the team consider real payments, institutional verification, or AI-powered image matching.

Suggested listing record

FIELD	WHY IT MATTERS
Title and category	Makes an item findable and keeps search results consistent.
Condition and photos	Sets buyer expectations and supports a clear handover checklist.
Exchange mode	Indicates purchase, daily/weekly rent, borrow, barter, donation, or free pickup.
Price, deposit, or budget	Separates seller price from refundable deposit and buyer request budget.
Availability and return date	Supports rental windows, booking conflicts, and return reminders.
Campus block or pickup zone	Enables campus-specific proximity without exposing unnecessary precise location.
Owner, status, and reports	Supports moderation, listing expiry, transaction state, and abuse follow-up.

Safety and truthfulness

A campus focus can reduce friction but does not make a marketplace “scam-free.” A production service needs reporting and moderation, careful verification, privacy controls, item-condition evidence, safe handover guidance, and clear rules for disputes. Lost-and-found matching must be presented as a suggestion rather than a guaranteed identification. Sustainability and savings values should be labelled estimates unless supported by a documented methodology.

03 / PRODUCT CONCEPT

OrbitX: AI-Driven Campus Fitness & Health

Live product website: https://xorbit-ruby.vercel.app/

The problem

Academic pressure can disrupt movement, eating, sleep, and recovery. Conventional routines may be difficult to follow when a student has limited equipment, unpredictable class schedules, budget constraints, crowded facilities, or little coaching support. A disconnected collection of fitness trackers can make the experience feel fragmented.

The solution direction

OrbitX is intended as a connected student health and fitness experience. It brings training, food awareness, movement, sleep, recovery, and habit-building into a single system that can adapt to a person's goals and available time.

Core capability map

CAPABILITY	INTENDED FUNCTION	RESPONSIBLE DESIGN NOTE
AI personal trainer	Suggest routines based on goals, equipment, experience, and schedule.	Ask about injuries or constraints responsibly; do not pretend a generic model can diagnose.
Exercise guidance	Give cues, substitutions, and progressions for exercises.	Use conservative guidance and advise qualified supervision when appropriate.
Food and macro tracker	Log meals and help interpret nutrition patterns.	Estimates vary by portion and data quality; avoid claims of exact nutritional measurement.
Budget-aware food ideas	Suggest balanced meals compatible with campus dining or common low-cost ingredients.	Respect personal preferences and dietary needs without moralizing food choices.
Activity and habits	Encourage achievable movement, hydration, and consistency milestones.	Prompts should support agency, not shame people for missed days.
Sleep and recovery	Help people notice patterns across rest, activity, and training load.	Signals are educational indicators, not medical conclusions.

What makes it student-aware?

- Time-bounded routines for busy timetables, exam weeks, and short gym visits.
- Equipment-aware alternatives that make a plan useful even when a machine or free-weight station is busy.
- Budget-sensitive nutrition ideas based on campus food options rather than idealized meal plans.
- Habit loops that balance activity, sleep, and recovery instead of optimizing only intensity.
- Progress displays that explain trends without claiming medical precision or guaranteed outcomes.

AI and health boundaries

A production health product requires explicit consent, data minimization, secure storage, a clear deletion/export process, careful model evaluation, and clinical review when features make health-related claims. It should avoid diagnosing conditions, instructing users to ignore clinicians, or using engagement mechanics that encourage harmful overtraining or restrictive eating. The front-end company site only describes the concept; it is not itself a clinical tool.

04 / THE WEBSITE

Information architecture

The SYNEQ page is a single responsive company website. Sections are stacked vertically, rather than placed in sideways sliders, so visitors can scroll naturally on desktop and mobile.

SECTION	VISITOR QUESTION	INTERACTION
Hero and ecosystem map	What is SYNEQ?	Product links, guided tour, welcome note after local profile save.
Product suite	What are the two products?	Cards with summary, capability tags, domain, and external launch links.
Capabilities	What does each product include?	Accessible tab control swaps UniTrade and OrbitX feature panels.
SYNEQ difference	How do we think about product design?	Animated principle cards and small visual illustrations.
Impact framework	What outcomes matter?	Outcome dashboard; clearly labels values as goals rather than verified results.
About	Why does the studio exist?	Objective, guiding principles, design philosophy.
Contributors	Who built the concept?	Photo, bio, skills, LinkedIn and GitHub links; missing LinkedIn is explicitly disabled.
Field notes / control room	What can the visitor do?	Theme, tour, screenshot, JSON export, personal PDF, static project guide.
Footer	Where else can I go?	Shortcuts into products, sections, and guide.

Motion and interaction system

- Startup sequence with progress messaging, a skip action, and a reduced-motion alternative.
- Scroll reveal transitions, a reading-progress bar, ambient gradients, orbit loops, floating cards, animated data bars, and hover states.
- Light/dark theme switch with persistence in localStorage.
- Quick search with filtered commands and keyboard navigation.
- Full-site tour with ten steps, progress, back/next controls, and a target spotlight.
- Floating quick-action dock for search, shortcuts, guided tour, and back-to-top.
- Responsive layout, keyboard focus, semantic page landmarks, and reduced-motion CSS.

Visual direction

The page uses a dark near-black foundation, warm orange highlights, cyan and violet support colours, gradient typography, thin technical rules, rounded product cards, simulated data visualizations, generous spacing, and quiet monospace labels. The light mode changes the page surfaces and foregrounds while retaining the product accent palette.

05 / FRONT-END ARCHITECTURE

Three source files, no build tool

The deliverable is intentionally static: index.html, style.css, and script.js. Open index.html directly in a modern browser or serve the folder through a local static server. No package installation, React, TypeScript, backend, or API key is required for the base site.

FILE	RESPONSIBILITY	KEY AREAS
index.html	Semantic structure and content.	Header and nav, hero, product cards, capability tabs, about, contributors, utility actions, dialogs, search panel, guided tour, toast region.
style.css	Visual system, responsive rules, motion.	CSS variables, dark/light themes, glass surfaces, animated illustrations, hover/reveal states, mobile breakpoints, reduced-motion, print styles.
script.js	Local interactions and exports.	Theme and profile persistence, nav, reveal observer, tabs, tips, search, keyboard shortcuts, guided tour, screenshot, JSON export, browser-generated PDF.
SYNEQ-Project-Guide.pdf	Long-form project explanation.	Concept, user problems, product features, architecture, UX, privacy, accessibility, production roadmap.

Run from scratch

1. Keep index.html, style.css, script.js, and SYNEQ-Project-Guide.pdf in the same folder. 2. Open index.html in a current desktop or mobile browser. 3. For more consistent browser behavior, run a static local server from that folder. 4. Use the product links to open the live external applications. 5. Test theme/profile persistence, search, tabs, tour, screenshot, JSON export, and both PDF downloads.

Example local static server commands

```
python3 -m http.server 8000
Then open http://localhost:8000 in the browser.
```

External browser libraries

The personalized PDF button uses jsPDF from a public CDN. Screenshot capture uses html2canvas from a public CDN. Google Fonts are also loaded remotely. These conveniences require an internet connection; if a CDN is blocked or unavailable, core browsing and product links still work, but the dependent export action can fail. The downloadable static guide remains a regular local PDF file.

Local-first behavior

The local profile is a display-name personalization, not a real user identity. The page stores only name, optional email, theme, current tip index, and tour-completion marker in localStorage. The export downloads a JSON object containing product summaries and prototype notes, and the personalized PDF is built inside the browser.

06 / EXPORTS AND DATA

What the controls do

CONTROL	OUTPUT	IMPORTANT DETAIL
Theme switch	Updates CSS theme and meta theme colour.	Persists as syneq_theme in localStorage.
Local profile	Display name plus optional email.	Persists as syneq_profile. No password is requested or stored.
Full-page screenshot	PNG captured from the main content area.	html2canvas and cross-origin images can fail due to browser or network restrictions.
Export page data	JSON file with products, capabilities, preference state, objective, principles, and caveats.	Contains locally entered profile name/email if present; handle exported files responsibly.
Personal PDF	Named PDF summary of the current profile, product directions, principles, and prototype boundaries.	Generated in the browser using jsPDF. No document is uploaded to a server.
Project guide download	Static SYNEQ-Project-Guide.pdf file.	Keep the PDF beside index.html so the link resolves.

Browser storage model

The implementation uses guarded localStorage helpers. Reads are parsed inside try/catch, and storage failures result in an explanatory toast rather than a hard crash. Example keys include syneq_theme, syneq_profile, syneq_tour_completed, and syneq_tip_index. Data remains in the current browser profile until cleared by the user or the browser; this is not cloud synchronization.

Data lifecycle recommendations

- Avoid adding passwords, student-ID photos, medical records, payment details, or other sensitive information to localStorage.
- If account features are added later, replace demo profile persistence with an established authentication service and server-side authorization.
- Document what is exported by the JSON and PDF controls and avoid accidentally including private data.
- Add a clear profile-removal function and, for a bigger application, provide a full account-data export and deletion workflow.
- Do not treat client-side fields, local counters, or browser state as trusted proof of identity or payment status.

07 / RESPONSIBILITY

Accessibility, privacy, and safety

Accessibility and motion

- Semantic HTML provides header, navigation, main, section, article, and footer landmarks.
- Keyboard users can reach links and controls; focus-visible outlines highlight the active control.
- The main navigation collapses into a mobile menu, and the feature tabs support left/right arrow navigation.
- The guided tour has back, next, finish, and close controls and announces progress through visible text.
- Reduced-motion preferences disable or shorten animation and reveal transitions.
- Interactive targets have labels; buttons are used for actions and links for navigation.

Privacy

The page does not connect the local display profile to a server. An email field is optional and is not verified. A localStorage record is editable by the user and can be read by scripts running on the same origin. For these reasons, this should never be described as secure authentication. For deployment, use HTTPS, limit third-party scripts, define a retention policy, and run a privacy review before collecting personal data.

Safety for UniTrade

- Use reporting, moderation queues, clear prohibited-item policies, and responsible identity verification.
- Encourage safe, public handovers and inspection of items before a transaction.
- Use compliant payment integration for real transactions; do not trust a front-end amount or a simulated split payment.
- Show AI match scores as suggestions and require human confirmation for lost-and-found matches.
- Label price estimates, savings, Karma points, and environmental values according to how they were calculated.

Safety for OrbitX

- Avoid presenting exercise, nutrition, sleep, or recovery outputs as diagnosis or professional medical care.
- Make goal setting and activity prompts supportive rather than coercive or shaming.
- Use qualified review for any clinical claims, higher-risk exercise instructions, and recommendations involving health conditions.
- Protect health and activity records with access control, encryption, consent, audit logging, and retention/deletion controls if the product is expanded.

Truthful presentation

The SYNEQ page labels impact values as goals, describes the account as local-only, and cautions that product descriptions are intended directions. This distinction is essential: a polished interface must not imply that a prototype has a verified backend, complete AI models, verified users, or independently measured outcomes.

08 / FROM PROTOTYPE TO PRODUCTION

A practical roadmap

PHASE	DELIVERABLES	VALIDATION GATE
1. Front-end experience	Responsive company site, product navigation, theme, tour, search, static guide, local exports.	Keyboard and mobile checks; no broken links or console errors; reduced-motion pass.
2. User research	Test major student workflows with representative students; document pain points and usability issues.	Observed task completion and prioritized feedback, not only visual approval.
3. UniTrade MVP backend	Authentication, listings, search, request flows, moderation, reporting, basic messaging, authorized uploads.	Authorization tests, privacy review, abuse cases, safety and dispute handling.
4. OrbitX MVP backend	Explicit profile consent, training preferences, meal/activity logs, routine history, safe recommendation boundaries.	Domain review, evidence checks, secure data handling, evaluation of unwanted recommendations.
5. Trust and payments	Production identity verification, real payment provider, audit trail, receipts, refunds, cancellation and dispute workflow.	Security review, legal/payment compliance, fraud and failure testing.
6. AI/analytics validation	Dataset governance, evaluations, explainability, model monitoring, transparent impact calculation.	Documented accuracy and limitations, human escalation, model and metrics review.

Technical hardening checklist

- Move trusted user state and authorization to a server; never rely on localStorage for access control.
- Use a proper database schema, server-side validation, rate limits, secure sessions, and audit logging.
- Add error reporting and automated tests for search, tabs, modals, profile handling, and exports.
- Test screen readers, keyboard-only operation, large text, mobile overflow, and common browser differences.
- Set a Content Security Policy and review third-party CDN dependencies before public launch.
- Create a retention policy, incident response plan, backup strategy, and a clear support path.
- Measure impact with published assumptions and reproducible calculations rather than placeholder counters.

Acceptance test checklist

CHECK	EXPECTED RESULT
Open or refresh page	A brief animated boot experience appears and can be skipped.
Switch theme and refresh	The selected theme is restored from this browser.
Save a display name	The header and welcome note update; name remains after refresh.
Log out	The local profile is removed, theme preference remains.
Search with /	Quick-find opens; arrows change selection; Enter opens the selected action.
Press G, T, P, Shift+S, ?	Tour, theme, profile, screenshot, and shortcut dialog actions trigger outside text fields.
Use feature tabs	Only the selected product panel is shown and announced.
Download outputs	Static guide, personal PDF, JSON, and PNG are named and saved as appropriate.
Enable reduced motion	Ambient animations and scrolling transitions are reduced.

09 / HANDOFF NOTES

Launch readiness and maintenance

Before making SYNEQ or either product public-facing at production scale, review the following handoff items. These steps are intentionally separate from visual polish: an impressive user interface is only one layer of a dependable product.

Release readiness

- Confirm every internal anchor and external product/profile URL works on desktop and mobile.
- Test the site with keyboard-only navigation, visible focus, reduced-motion enabled, 200% zoom, and a narrow mobile viewport.
- Verify that the profile name persists on refresh, logout removes only the profile, and theme preference remains intact.
- Test screenshot and PDF exports with the contributor images loaded and blocked; make error messages understandable.
- Keep the companion project guide in the same directory as index.html and check the download link after hosting.
- Review all public metrics, bios, launch dates, product features, and trust/AI statements for accuracy before publishing.

Maintenance checklist

CADENCE	TASKS
Every release	Check navigation, links, responsive overflow, console errors, theme, profile, search, tabs, and tour.
When copy changes	Update section headings, search index records, tour descriptions, and the PDF guide so they stay aligned.
When a third-party library changes	Review version, CDN availability, dependency integrity, privacy impact, and fallback behavior.
Before real user data	Threat-model storage, confirm consent and retention, introduce server-side access control, and review data export/deletion.
Before AI claims	Document evaluation, known failure modes, human escalation, and how generated recommendations are explained.

Short glossary

TERM	PLAIN-LANGUAGE MEANING
LocalStorage	Browser storage that can persist small key/value data for a site in the same browser profile. It is not a secure database.
Responsive layout	A layout that adapts to different screen widths and input methods.
Reduced motion	A system preference asking interfaces to limit movement and animation.
Prototype metric	A value used to illustrate a goal or user interface, not necessarily a verified result.
Human confirmation	A person reviews an AI suggestion before it becomes a consequential decision.

Closing note

SYNEQ is designed as a clear, ambitious front door to two useful student-focused product concepts. The next step is not simply to add more animation; it is to validate the underlying workflows with students, improve reliability and accessibility, and implement secure infrastructure where real accounts, health information, payments, and trust decisions are involved.